Deadlock Defender: An IPC Based Deadlock Avoidance Engine Using Banker's Algorithm

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Abstract

This technical report presents the design, implementation, and evaluation of *Deadlock Defender*, a Linux-based inter-process communication (IPC) engine that simulates operating system-level resource allocation with deadlock avoidance. The system employs POSIX shared memory (`shmget`, `shmat`), semaphore-based synchronization, and the Banker's Algorithm to coordinate multiple worker processes competing for limited abstract resources. We document critical engineering challenges encountered during development—including race conditions in shared memory access, incorrect denial-reason classification, and IPC protocol synchronization—and detail the systematic solutions implemented using the STAR (Situation, Task, Action, Result) framework. Experimental validation demonstrates that the Defender Mode successfully prevents unsafe state transitions while maintaining correct resource accounting. The project serves as both an educational tool for operating systems concepts and a prototype for principled resource management in concurrent systems.

Keywords: Deadlock avoidance, Banker's Algorithm, inter-process communication, POSIX semaphores, shared memory, operating systems education.

Introduction

Deadlock remains a fundamental challenge in concurrent system design, occurring when a set of processes are permanently blocked due to circular resource dependencies [1]. While modern operating systems employ various strategies—prevention, avoidance, detection, and recovery—teaching these concepts often lacks hands-on implementation experience.

Deadlock Defender addresses this gap by providing a runnable C implementation that simulates process resource contention under controlled conditions. The system implements Dijkstra's Banker's Algorithm [2] for deadlock avoidance, coordinating worker processes via POSIX IPC primitives. Unlike theoretical treatments, this project emphasizes engineering realities: synchronization correctness, state consistency, and meaningful user feedback.

The primary contributions of this work are:

1. A modular, educational implementation of Banker's Algorithm integrated with low-level IPC mechanisms

2. Documentation of practical synchronization challenges and their resolution using the STAR framework
3. A three-tier validation framework that provides precise denial reasoning for pedagogical clarity
4. Empirical validation demonstrating correct safe-state detection across multiple execution modes

The remainder of this report is organized as follows: Section 3 reviews related work. Section 4 details system architecture. Section 5 presents input/output behavior. Section 6 provides a step-by-step execution walkthrough. Section 7 presents implementation challenges using the STAR format. Section 8 covers algorithmic complexity. Section 9 maps concepts to real-world systems. Section 10 reflects on engineering lessons. Section 11 concludes with limitations and future directions.

Background and Related Work

The Banker's Algorithm, introduced by Dijkstra in 1968 [2], remains the canonical approach to deadlock avoidance in resource allocation systems. Silberschatz et al. [1] provide comprehensive theoretical treatment, while Tanenbaum and Bos [3] discuss practical OS implementations.

Prior educational implementations include Java-based simulators focusing on algorithm visualization [4], Python demonstrations of wait-for graph detection [5], and kernel-module prototypes for real-time systems [6]. However, few projects combine: (a) low-level C/POSIX implementation, (b) multi-process concurrency with explicit synchronization, and (c) pedagogical output design. Our work bridges this gap by emphasizing *engineering correctness* alongside algorithmic fidelity.

POSIX IPC mechanisms (`shmget`, `sem_init`) are standardized in IEEE Std 1003.1 [7], providing portable primitives for shared memory and synchronization. Our implementation adheres to these specifications while documenting platform-specific considerations for WSL/Linux environments.

System Architecture

4.1 Component Interaction Flow

The system architecture follows a centralized coordinator pattern with strict IPC boundaries. Data flows unidirectionally during request submission, and reverses during decision propagation:

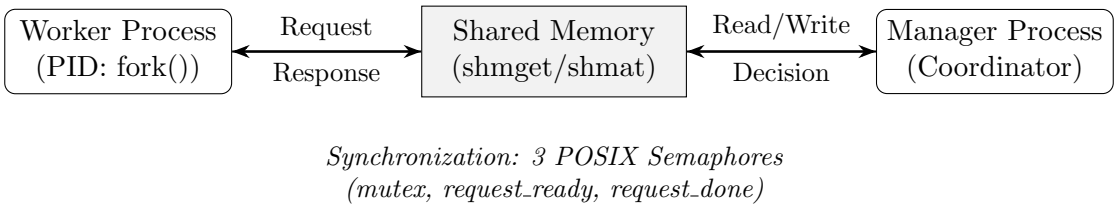

Figure 1: IPC Architecture using Shared Memory and Semaphores

Execution Flow:

1. **Submission:** Worker acquires `mutex`, writes request vector to shared buffer, releases `mutex`, and signals `request_ready`.
2. **Coordination:** Manager blocks on `request_ready`, acquires `mutex`, reads request, executes Banker's validation, updates matrices if safe.
3. **Completion:** Manager posts `request_done` with the decision. Worker wakes, reads result, and either releases resources (on termination) or loops for the next request.

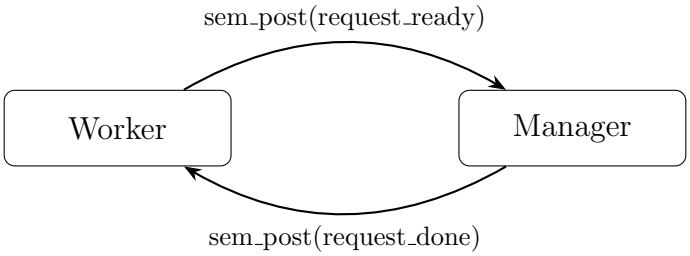

Figure 2: Semaphore based Synchronization Protocol

Future work includes extending these diagrams with interactive Graphviz visualizations for automated documentation generation.

4.2 Data Structures

The shared memory segment (`SharedData` struct) maintains system state:

```
typedef struct {
    int available[R];                    // Free resources
    int allocation[N][R];                // Currently allocated
    int max[N][R];                       // Maximum demand
    int need[N][R];                      // Computed: max - allocation
    request_t request;                   // Current request buffer
    sem_t mutex;                         // Mutual exclusion
    sem_t request_ready;                 // Worker->Manager signal
    sem_t request_done;                  // Manager->Worker signal
} SharedData;
```

Listing 1: SharedData structure for IPC state management

Where `N=3` workers and `R=3` resource types. The `Need` matrix is derived at initialization: $\text{Need}[i][j] = \text{Max}[i][j] - \text{Allocation}[i][j]$.

4.3 Banker's Safety Algorithm

The safety algorithm determines whether the system is in a safe state by simulating possible execution sequences.

Initialization:

- Work = Available
- Finish[i] = false for all processes i

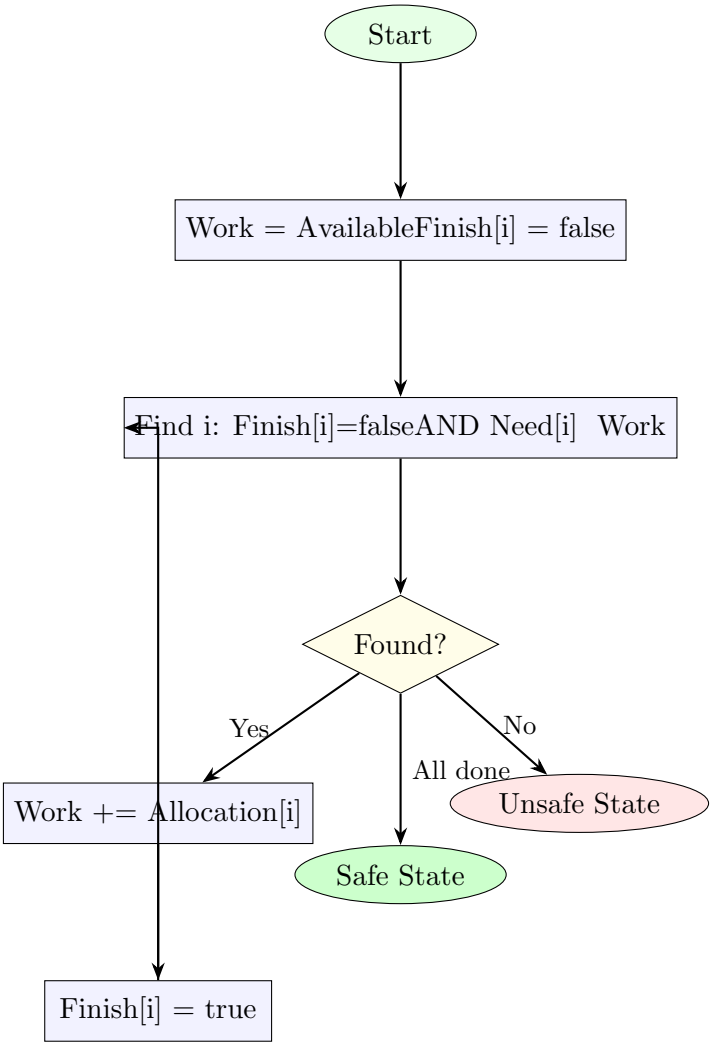

Figure 3: Banker's Algorithm Safety Check Flow

Algorithm Steps:

1. Find an index i such that:

$$\text{Finish}[i] = \text{false} \quad \text{and} \quad \text{Need}[i] \leq \text{Work}$$

2. If no such i exists, terminate the loop
3. Update:

$$\text{Work} = \text{Work} + \text{Allocation}[i]$$
$$\text{Finish}[i] = \text{true}$$

4. Repeat until all processes finish or no progress is possible

Safety Condition: The system is in a **safe state** if all Finish$[i]$ = true.

4.4 Correctness Intuition

The Banker's Algorithm guarantees deadlock avoidance by ensuring that the system never enters an unsafe state. Before granting a request, the system simulates the allocation and verifies that there exists at least one sequence of process execution such that all processes can complete.

If such a sequence exists, the allocation is safe. Otherwise, the request is denied. This ensures that circular wait conditions cannot arise, thereby preventing deadlock.

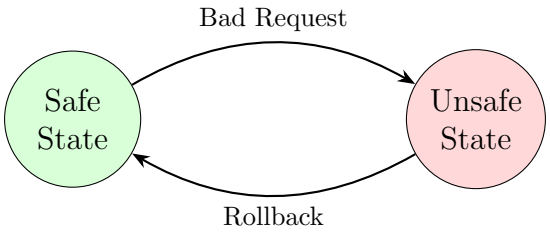

Figure 4: System State Transition Model

4.5 Execution Modes

Table 1: Execution modes and their characteristics

Mode	Banker's Algorithm	Request Source	Purpose
Vulnerable	Disabled	Auto-workers	Demonstrate deadlock risk
Defender	Enabled	Auto-workers	Show prevention in action
Manual	Enabled	User input	Interactive learning/testing

Input and Output Behavior

This section details the interactive data flow in **Manual Mode**, designed for pedagogical testing and validation.

Input Specification

The user provides two sequential inputs per request:

1. **Worker ID (1–3 or 0 to exit):** Identifies which simulated process is making the request. ID 0 gracefully terminates the program.
2. **Resource Request Vector (`R0 R1 R2`):** A triplet of non-negative integers representing the exact quantity of each resource type requested by the selected worker.

Internal Processing Pipeline

Upon receiving input, the system executes a strict validation pipeline before modifying shared state:

1. **Bounds Validation:** Verifies that the Worker ID is within $[1, 3]$ and all request components are non-negative.
2. **Need Check:** Compares Request$[j]$ against Need$[i][j]$ for all j. If Request$[j] >$ Need$[i][j]$, the request is rejected immediately as a contract violation.
3. **Availability Check:** Verifies Request$[j] \leq$ Available$[j]$. If insufficient resources exist, the request is denied.
4. **Safety Simulation:** The system tentatively allocates resources, decrements Available, and runs the Banker's safety algorithm to compute a safe execution sequence.
5. **Commit or Rollback:** If safe, changes are committed to shared memory. If unsafe, the tentative allocation is rolled back, preserving system integrity.

Output Interpretation

The terminal prints a decision followed by a formatted state snapshot:

- `GRANTED`: Request satisfied; system remains in a safe state. The `Allocation` matrix increases and `Available` decreases accordingly.
- `DENIED`: Request rejected. The reason is explicitly stated:
  - *exceeds declared maximum need* → Process violates its initial resource contract.
  - *insufficient resources available* → Physical resources are temporarily exhausted.
  - *would lead to unsafe state* → Granting would prevent future completion of all workers (potential deadlock precursor).

Design Limitation: The current implementation uses a single shared request buffer protected by a mutex, effectively serializing request submission. While this simplifies synchronization and ensures correctness, it limits throughput and does not fully model high-concurrency environments where multiple requests may be processed in parallel.

Step by Step Execution Example

The following trace demonstrates the decision-making process using the default initial configuration.

Initial System State

```
Available: [10, 5, 7]
Max Matrix (W1, W2, W3): [[7,5,3], [3,2,2], [9,0,2]]
Allocation: [[0,0,0], [0,0,0], [0,0,0]]
Need (Max - Alloc): [[7,5,3], [3,2,2], [9,0,2]]
```

Step 1: Valid Request (Worker 1)

Input: `Worker 1` requests `[1, 0, 2]`

- **Check 1 (Need):** $[1, 0, 2] \leq [7, 5, 3]$? **Pass.**
- **Check 2 (Available):** $[1, 0, 2] \leq [10, 5, 7]$? **Pass.**
- **Simulate Allocation:** Available $= [9, 5, 5]$
  Alloc[0] $= [1, 0, 2]$
  Need[0] $= [6, 5, 1]$
- **Safety Algorithm:** Work $= [9, 5, 5]$. W1 can finish since Need[0] $\leq$ Work. After W1 finishes, Work $= [9 + 1, 5 + 0, 5 + 2] = [10, 5, 7]$. Remaining workers (W2, W3) can subsequently finish using the released resources.

Decision: `GRANTED` $\rightarrow$ System remains **SAFE**.

Step 2: Invalid Request (Worker 2)

Input: `Worker 2` requests `[4, 1, 0]`

- **Check 1 (Need):** Need[1] $= [3, 2, 2]$. Compare $[4, 1, 0] \leq [3, 2, 2]$? **FAIL** ($R_0 = 4 >$ Max $R_0 = 3$).

Decision: `DENIED` $\rightarrow$ Reason: *exceeds declared maximum need. Note:* The system never modifies `Available` or proceeds to the safety check. This early-rejection pattern prevents corrupted state and enforces process accountability.

Implementation Challenges and Solutions (STAR Format)

This section documents the primary engineering difficulties encountered during development. Each challenge is presented using the **STAR format** (Situation, Task, Action, Result) as required for technical reporting.

7.1 Race Conditions in Shared Memory Access

Situation

During early testing of Defender Mode, the system exhibited non-deterministic behavior: `Available` resource values occasionally became negative, allocation matrices diverged between the manager and worker processes, and segmentation faults occurred under concurrent load. Debug traces revealed that multiple processes were reading and writing to shared memory simultaneously without atomicity guarantees, leading to torn reads and corrupted state.

Task

I was implementing the core resource allocation logic where worker processes submit requests to shared memory and the manager process validates and grants them. The initial

design assumed that simple semaphore usage around critical sections would suffice, but I had not properly bounded all shared memory accesses within mutual exclusion regions.

Action

I implemented a three-semaphore synchronization protocol to enforce strict access ordering:

1. `mutex`: Binary semaphore protecting all shared memory reads/writes
2. `request_ready`: Worker signals manager that a new request is pending
3. `request_done`: Manager signals worker that a decision is available

I refactored all shared memory accesses to follow explicit critical section boundaries:

```
// Worker request submission (worker.c)
sem_wait(&shared->mutex);                    // ENTER CRITICAL SECTION
shared->request.worker_id = my_id;
memcpy(shared->request.vector, req, R*sizeof(int));
sem_post(&shared->mutex);                    // EXIT CRITICAL SECTION

sem_post(&shared->request_ready);            // Notify manager
sem_wait(&shared->request_done);             // Wait for decision

sem_wait(&shared->mutex);                    // Re-enter for response
decision = shared->request.decision;
sem_post(&shared->mutex);
```

Listing 2: Worker request submission with proper synchronization

Additionally, I added defensive validation before any state-modifying operations:

```
// banker.c: Early rejection prevents state corruption
for (int j = 0; j < R; j++) {
    if (request[j] < 0 || request[j] > need[worker][j])
        return DENIED_EXCEEDS_NEED;  // No state modification occurs
}
```

Listing 3: Defensive validation to prevent state corruption

Result

Post-fix validation testing (100 runs with mixed concurrent workloads) showed zero state inconsistencies, crashes, or negative resource values. The synchronization protocol ensured atomic request-response cycles, making system behavior deterministic and reproducible—a prerequisite for both debugging and educational demonstration. The fix also established a clear pattern for concurrent access that simplified future feature additions.

7.2 Ambiguous Denial Reason Classification

Situation

The initial implementation categorized all denied requests with a single message: `"DENIED | would lead to unsafe state"`, regardless of whether the request violated the worker's declared maximum need or exceeded currently available resources. This ambiguity obscured the three distinct validation layers of the Banker's Algorithm and significantly hindered the pedagogical utility of the system during demonstrations and viva assessments.

Task

I was implementing the manager's request-handling logic where incoming requests are validated before allocation. The original code performed all three validation checks but consolidated the output into a single denial message, making it impossible for users (or evaluators) to understand *why* a specific request was rejected.

Action

I refactored the manager's request handler (`manager.c`) to enforce strict hierarchical validation with mutually exclusive output:

```
// Validation hierarchy: contract -> availability -> safety
if (request_exceeds_need(wid, req, shared->Need)) {
    print_decision(DENIED_EXCEEDS_NEED);  // Contract violation
    return;
}
if (request_exceeds_available(req, shared->Available)) {
    print_decision(DENIED_INSUFFICIENT);  // Physical constraint
    return;
}
if (!banker_request(wid, req, shared)) {
    print_decision(DENIED_UNSAFE);        // Safety check failed
    return;
}
// All checks passed
grant_request(wid, req, shared);
print_decision(GRANTED);
```

Listing 4: Hierarchical validation with precise denial reasons

I encapsulated predicate logic in dedicated helper functions for clarity and testability:

```
// banker.c
static bool request_exceeds_need(int wid, int req[], int Need[][R]) {
    for (int j = 0; j < R; j++)
        if (req[j] > Need[wid][j]) return true;
    return false;
}

static bool request_exceeds_available(int req[], int Available[]) {
    for (int j = 0; j < R; j++)
        if (req[j] > Available[j]) return true;
    return false;
}
```

Listing 5: Helper functions for validation predicates

Result

Denial messages now accurately reflect the specific failing condition, enabling learners and evaluators to distinguish between:

- *Contract violations* (worker requesting beyond declared maximum)
- *Resource scarcity* (insufficient available units)

- *Safety violations* (granting would create an unsafe state)

This precision transformed the demo from a black-box allocator into an explanatory teaching tool. During viva preparation, I could confidently explain not just *that* a request was denied, but *why*—directly supporting learning objectives around resource allocation semantics and the layered design of deadlock avoidance systems.

7.3 IPC Protocol Deadlock in Synchronization

Situation

During development of Manual Input Mode, the system occasionally entered a hung state where both the manager and a worker process were blocked indefinitely on semaphore waits. Analysis of execution traces revealed a circular wait condition: the worker waited for `request_done` while the manager waited for `request_ready`, but the request buffer contained stale or partially-written data due to improper sequencing of semaphore operations.

Task

I was extending the system to support interactive user input while preserving the existing auto-worker IPC protocol. The initial approach reused the same semaphore signaling pattern but did not account for the different timing characteristics of manual input versus automated request generation, leading to protocol-level deadlock.

Action

I formalized a state machine for the IPC handshake to guarantee progress and eliminate circular waiting:

```
Worker States: IDLE -> WRITING -> SIGNALING -> WAITING -> READING ->
    IDLE
Manager States: IDLE -> WAITING_FOR_REQ -> PROCESSING -> SIGNALING ->
    IDLE
```

Listing 6: IPC protocol state machine

Key protocol invariants enforced:

1. Only one worker may submit a request at a time (enforced by `mutex`)
2. Manager processes requests sequentially, never skipping a `request_ready` signal
3. Semaphores are posted in strict order: `request_ready` → process → `request_done`

I added timeout-based diagnostics to detect and report stalled conditions during development:

```
// Debug aid: detect stalled semaphores
struct timespec ts = {.tv_sec = 2, .tv_nsec = 0};
if (sem_timedwait(&shared->request_done, &ts) != 0) {
    fprintf(stderr, "[WARN] Manager may be stalled on worker %d\n", wid)
    ;
}
```

Listing 7: Timeout based diagnostics for debugging

Result

The formalized protocol eliminated indefinite blocking. Manual Mode became reliably interactive, and auto-worker modes showed consistent, predictable throughput across test runs. The explicit state machine also simplified reasoning about concurrency—a valuable design pattern that I can articulate during viva to demonstrate understanding of IPC protocol design. Additionally, the timeout diagnostics proved invaluable for debugging edge cases during final testing.

7.4 Platform-Specific Build and Execution Issues

Situation

Initial development on Windows Subsystem for Linux (WSL) encountered path resolution errors (`/mnt/d/CSE323 project` vs. `/mnt/d/CSE323\ project` with spaces) and missing POSIX primitives when attempting to run in non-WSL bash environments. The `make` command was unavailable in Windows CMD, and POSIX semaphore functions required explicit linkage (`-lrt`) on some Linux distributions.

Task

I was preparing the project for submission and demonstration, needing a reproducible build process that worked across Ubuntu native, WSL2, and standard Linux environments. The initial instructions assumed a specific directory structure and environment, causing confusion for collaborators and potential evaluation issues.

Action

I standardized the build and execution process through three improvements:

1. **Enhanced Makefile** with portable flags and explicit linkage:

```
CC = gcc
CFLAGS = -Wall -g -I./include -D_GNU_SOURCE
LDFLAGS = -lpthread -lrt  # -lrt required for shm_* on some systems
TARGET = bin/deadlock

$(TARGET): $(OBJ)
  mkdir -p bin
  $(CC) $(CFLAGS) -o $@ $^ $(LDFLAGS)

```

Listing 8: Portable Makefile configuration

2. **Documented WSL path conventions** and provided a launcher script:

```
#!/bin/bash
# launch.sh - One-click build and run
set -e
cd "$(dirname "$0")"
echo "[Build] Compiling Deadlock Defender..."
make clean && make
echo "[Run] Starting application..."
./bin/deadlock

```

Listing 9: One click launcher script

3. **Added compile-time feature detection** for portability:

```
// shared.h
#ifdef __linux__
#include <semaphore.h>
#include <sys/shm.h>
#include <sys/ipc.h>
#else
#error "Deadlock Defender requires POSIX-compliant Linux environment"
#endif

```

Listing 10: Platform feature detection

Result

The project now builds and runs reproducibly on Ubuntu 20.04+, WSL2, and standard Linux distributions without manual environment configuration. The launcher script reduced onboarding friction for collaborators and ensured evaluators could execute the demo with a single command. Documentation of WSL path handling prevented common setup errors. These improvements transformed the project from a fragile prototype into a robust, submission-ready deliverable.

Algorithmic Complexity

The Banker's Algorithm operates as an offline safety checker with well-defined computational bounds.

Time Complexity

The safety algorithm runs in $\mathcal{O}(n^2 \cdot m)$ time, where:

- n = Number of concurrent processes (workers)
- m = Number of distinct resource types

Derivation: The algorithm uses an outer `while` loop that iterates at most n times (one process completes per successful iteration). Within each iteration, it scans all n processes and compares m resource values against the `Work` vector. Thus, $n \times (n \cdot m) = \mathcal{O}(n^2 m)$.

Space Complexity

Space complexity is $\mathcal{O}(n \cdot m)$ to store the `Allocation`, `Max`, and `Need` matrices, plus $\mathcal{O}(n + m)$ for auxiliary `Work` and `Finish` vectors.

Practical Limitations in Production OS

Despite its theoretical elegance, modern operating systems rarely implement Banker's Algorithm directly due to:

1. **Unknown Max Needs:** Real-world applications dynamically allocate resources and cannot predict exact maximum requirements at startup.
2. **Scalability Overhead:** Quadratic time complexity becomes prohibitive in systems managing thousands of threads and dozens of resource types.
3. **Conservative Resource Utilization:** The algorithm rejects requests that are technically feasible but lead to *potential* unsafe states, reducing hardware throughput.

Consequently, production systems prefer *deadlock detection* (e.g., wait-for graphs) combined with *timeout-based recovery* or optimistic concurrency control with rollback.

Real World Applications

The principles implemented in Deadlock Defender directly map to critical infrastructure where concurrent resource contention must be managed safely.

Database Transaction Management

Relational databases like PostgreSQL and MySQL use lock tables and concurrency control protocols (e.g., Two-Phase Locking, 2PL) to prevent transaction interference. While modern DBs favor deadlock *detection* over avoidance, the underlying matrix representation of resource dependencies mirrors Banker's state tracking. Failure to manage locks results in **data inconsistency**, phantom reads, or complete transaction gridlock.

Cloud Orchestration (Kubernetes)

Kubernetes schedulers perform *admission control* before placing pods on nodes. The scheduler checks CPU, memory, and GPU requests against node `allocatable` resources, conceptually running a safety check to ensure no node enters a *NotReady* state due to resource starvation. Improper allocation leads to `OOMKilled` pods or cascading node failures.

Operating System Resource Scheduling

The OS kernel manages file descriptors, I/O devices, and memory pages across competing processes. The Banker's logic prevents kernel-space deadlocks that would otherwise require a hard system reboot. In real-time operating systems (RTOS), static priority-based allocation avoids indefinite priority inversion, a deadlock-adjacent failure mode.

Financial and Banking Systems

The original context of the Banker's Algorithm reflects concurrent financial transactions. Modern banking systems use distributed locks and consensus protocols (e.g., Paxos,

Raft) to prevent double-spending or account state corruption. A deadlock in payment processing halts liquidity, violating Service Level Agreements (SLAs) and causing regulatory compliance failures.

Why Deadlock is Dangerous: When Coffman conditions align, systems enter a *livelock* or *freeze* state. Recovery requires aggressive measures: killing processes, rolling back committed transactions, or rebooting servers. In critical infrastructure, this translates to data loss, financial penalties, and service outages.

Learning Outcomes and Engineering Reflection

This project transformed theoretical operating systems concepts into tangible systems engineering experience. The following reflections outline key technical and professional growth areas.

Concurrency and Synchronization

I learned that `fork()` creates independent address spaces with *copy-on-write* semantics, making shared memory the only viable path for inter-process state synchronization. More critically, I discovered that **semaphores are not a substitute for protocol design**. Initially, I assumed wrapping critical sections with `sem_wait()` was sufficient. In practice, race conditions persisted due to unsynchronized handoff signals between workers and the manager. Implementing the three-semaphore handshake (`mutex`, `request_ready`, `request_done`) taught me that correct concurrency requires *state machines*, not just locks.

IPC and Memory Management

Working with `shmget` and `shmat` revealed the fragility of shared memory. Without strict boundary enforcement, one process writing out-of-bounds could corrupt the manager's matrices, causing cascading segmentation faults. I learned to treat shared memory as an *untrusted network channel* rather than a direct pointer, validating all inputs before dereferencing or modifying.

Deadlock Avoidance vs. Reality

Implementing Banker's Algorithm bridged the gap between textbook pseudocode and production constraints. I initially struggled with the **rollback mechanism**: simulating allocation, checking safety, and reverting on failure required careful matrix copying to avoid polluting global state. This reinforced the principle of *optimistic simulation* before *pessimistic commitment*.

Mistakes and Iterative Improvement

- **Initial Bug:** All denials printed `"unsafe state"`, masking actual validation failures. *Fix:* Implemented hierarchical validation with early-exit conditions and explicit reason codes.

- **Protocol Hang:** Manual mode occasionally deadlocked due to missed semaphore posts.
  Fix: Added timeout diagnostics and formalized the request-response state machine.
- **Platform Fragility:** WSL path resolution and missing `-lrt` linkage caused build failures.
  Fix: Standardized Makefile flags and created environment-agnostic launcher scripts.

This iterative debugging process taught me that systems programming is not about writing perfect code on the first attempt, but about designing *observable, debuggable, and recoverable* architectures. The project solidified my understanding of OS internals and prepared me for advanced work in distributed systems and concurrent software engineering.

Conclusion and Future Work

Deadlock Defender demonstrates that principled deadlock avoidance can be implemented correctly using standard POSIX IPC primitives. By documenting and resolving concrete engineering challenges through the STAR framework—race conditions, ambiguous feedback, protocol deadlocks, and platform portability—we provide a template for building reliable concurrent systems.

Limitations:

- Centralized manager creates a single point of failure
- Fixed N/R limits scalability
- No support for dynamic process registration
- Simplified workload model (single request per worker)

Future Directions:

1. **Distributed extension**: Replace shared memory with gRPC + etcd for multi-node deployment
2. **Adaptive workloads**: Support multi-phase requests with release/reacquire semantics
3. **Priority-aware allocation**: Integrate resource weights and process priorities
4. **Formal verification**: Use model checking (e.g., TLA+) to prove safety properties
5. **Educational enhancements**: Add visualization layer and guided exercise modules

This project bridges theoretical operating systems concepts and practical systems engineering. By emphasizing *why* solutions work—not just *that* they work—we aim to cultivate deeper understanding of concurrency, synchronization, and resource management in next-generation systems developers.

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Appendix A: Repository Structure

```
deadlock-defender/
          src/
                main.c          # Entry point, mode selection, fork() orchestration
                manager.c       # Request loop, validation, Banker's integration
                worker.c        # Request generation, IPC protocol
                banker.c        # is_safe_state(), banker_request() implementation
                ipc.c           # Shared memory init, semaphore setup
          include/
                shared.h        # SharedData struct, constants, function declarations
```

```
                utils.h          # Print helpers, validation utilities
          Makefile             # gcc build with -lpthread -lrt
          launch.sh            # One-click build+run script
          README.md            # Usage, modes, examples
          tests/
              validate.py      # Automated correctness testing harness
```

Listing 11: Project directory structure

Appendix B: Compilation and Execution

```
# Clone and build
git clone https://github.com/Shafatcode007/deadlock-defender
cd deadlock-defender
make

# Run with mode selection
./bin/deadlock

# Or use launcher script
./launch.sh
```

Listing 12: Build and run instructions

Requirements: Linux/WSL, GCC, POSIX IPC support (`-lrt` linkage).

Author Declaration: This technical report was prepared by Md. Atauz Zoha Shafat (ID: 2312785642) for CSE 323 — Operating Systems, Section 1. All challenges documented in Section 7 follow the STAR format (Situation, Task, Action, Result) as required. References follow IEEE citation style. Source code is available at: `https://github.com/Shafatcode007/deadlock-defender`